

PAPER417 – π Cycles and Negative Time: $\cos(\pi t_n)$ Temporal Reversal Framework in UQFF

Session: 110 (grokshare755feea7.txt analysis)

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Source: grokshare755feea7.txt — "Star Magic" Chapter 8 + FU Temporal Modulation Sections **Session:** 110 (grokshare755feea7.txt analysis) **CP4 Class:** PiCyclesNegativeTimeCosineTemporalReversalCalculator (#67)

1. Overview

PAPER_417 derives and formalizes the $\cos(\pi t_n)$ temporal modulation factor throughout UQFF, where $t_n = t - t_0$ is a **shifted time variable** that admits negative values (time reversal relative to reference epoch t_0). This paper: (a) identifies all UQFF terms carrying this factor, (b) shows that negative t_n creates field reversals with physical consequences, (c) derives the non-linear oscillation factor in U_m , and (d) connects the π -cycle encoding to the Riemann Hypothesis prime distribution hypothesis.

2. The t_n Variable

2.1 Definition

$$t_n \equiv t - t_0$$

where t_0 is a **reference epoch** (e.g., stellar formation time, observer frame, or simulation start). For a star of age τ :

$$t_0 = 0: t_n = t \text{ (forward time from formation)}$$

$$t_0 = \tau: t_n = t - \tau \text{ — negative for } t < \tau \text{ (past events)}$$

$$t_0 = t: t_n = 0 \text{ — present moment}$$

2.2 Physical Range

$$t_n \in (-\infty, +\infty)$$

For astrophysical time: $t_n \in (-13.8 \text{ Gyr}, +t_{\text{observing}})$

3. $\cos(\pi t_n)$ Occurrences in UQFF

Term	Where $\cos(\pi t_n)$ appears	Effect
U_{g_1}	$\cos(\pi t_n)$ factor	Forward: constructive; $t_n < 0$: inverted DPM
U_{b_i}	$\cos(\pi t_n)$ factor	Forward: stabilizing buoyancy; past: destabilizing

Term	Where $\cos(\pi t_n)$ appears	Effect
U_m	$(1 - e^{-\gamma t} \cos(\pi t_n))$ exponent	Non-linear oscillation in magnetic strings
$A_{\mu\nu}$	t_n in $T_S^{\mu\nu}$ signature	Metric reversal for pre-formation epoch

3.1 Full U_{g1} with π Factor

$$U_{g1} = k_1 \cdot \mu_s(t, [\text{SCm}]) \cdot \nabla \left(\frac{M_s}{r} \right) \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + \delta_{\text{def}})$$

For $t_n = 0$: $\cos(0) = 1 \rightarrow$ maximum U_{g1} (present epoch) For $t_n = 1/2$: $\cos(\pi/2) = 0 \rightarrow U_{g1} = 0$ (null crossing) For $t_n = 1$: $\cos(\pi) = -1 \rightarrow U_{g1}$ inverted (one cycle prior = anti-epoch)

3.2 Full U_{bi} with π Factor

$$U_{bi} = -\beta_i \cdot U_{gi} \cdot \Omega_g \cdot \frac{M_{\text{BH}}}{d_g} \cdot (1 + \epsilon_{\text{sw}} \cdot \rho_{\text{sw}}) \cdot U_{\text{UA}} \cdot \cos(\pi t_n)$$

Negative t_n : $\cos(\pi t_n)$ changes sign $\rightarrow$ **buoyancy reversal** — the stabilizing term becomes destabilizing, offering a mechanism for pre-formation instabilities.

3.3 Non-Linear U_m Oscillation

$$U_m = \sum_j \frac{\mu_j(t, [\text{SCm}])}{r_j} \cdot \left(1 - e^{-\gamma t} \cos(\pi t_n) \right) \cdot \hat{\phi}_j \cdot P_{\text{SCm}} \cdot E_{\text{react}}$$

The exponent $- \gamma t \cdot \cos(\pi t_n)$ creates:

Normal epoch ($t_n > 0, \cos > 0$): Standard exponential saturation of U_m

Reversal epoch ($t_n < 0, \cos < 0$): $e^{+\gamma t |\cos(\pi t_n)|} \rightarrow U_m$ **grows exponentially** rather than saturating $\rightarrow$ quasar jet ignition mechanism

This exponential growth in U_m for negative t_n provides a pre-formation field amplification, consistent with observing highly magnetized quasars at early cosmic epochs.

4. Physical Consequences of Negative t_n

4.1 Field Reversal Table

t_n	$\cos(\pi t_n)$	U_{g1} behavior	U_m behavior	U_{bi} behavior
$t_n = 0$	+1	Maximum	Saturating	Stabilizing
$t_n = 0.5$	0	Zero	Frozen at current value	Zero
$t_n = 1$	-1	Inverted	Growing exponentially	Destabilizing
$t_n = -0.5$	0	Zero	Frozen	Zero
$t_n = -1$	-1	Inverted	Growing	Destabilizing

4.2 Quasar Asymmetric Jets (Revisited)

From PAPER_414, quasar jet asymmetry originates in $\cos(\pi t_n)$. With $t_n \neq 0$:

$$\frac{F_{j,+}}{F_{j,-}} = \frac{\cos(\pi t_n^{(+)})}{\cos(\pi t_n^{(-)})} \neq 1 \quad \text{when } t_n^{(+)} \neq -t_n^{(-)}$$

The **two opposing jets** correspond to two opposite t_n values — one being the advanced and one the retarded field, creating natural asymmetry without tuning.

5. Riemann Hypothesis Connection

The $\cos(\pi t_n)$ term introduces **oscillations at prime-like intervals** when t_n takes values associated with Riemann zeros $1/2 + i\gamma_k$:

```
\text{If } t_n = \text{Im}(\rho_k)/\pi \quad \text{(Riemann non-trivial zeros)}, \text{ then:} \\ \cos(\pi t_n) = \cos(\text{Im}(\rho_k)) \quad \text{→ UQFF field zeros at prime-counting events}
```

This is a hypothetical connection: the UQFF framework's π -cycle temporal modulations mirror the oscillatory structure of the prime counting function $\pi(x)$ error term through the explicit formula:

```
\pi(x) = \text{Li}(x) - \sum_{\rho} \text{Li}(x^{\rho}) + \dots
```

6. Code Implementation

```
// Temporal modulation in UQFF – main.cpp notation
double cos_pi_tn = cos(M_PI * tn);
// Ug1 modulation
double Ug1 = k1 * mu_s * grad_Ms_r * exp(-alpha * t) * cos_pi_tn * (1.0 + delta_def);
// Um non-linear modulation
double Um_factor = 1.0 - exp(-gamma * t * cos_pi_tn);
double Um = mu_j / r_j * Um_factor * P_Scm * Ereact;
// Ubi modulation
double Ubi = -beta_i * Ugi * Omega_g * Mbh / dg * (1.0 + eps_sw * rho_sw) * U-UA * cos_pi_tn;
```

7. Unit Tests

```
import math
def test_cos_pi_tn_symmetry():
    """cos(pi * tn) = cos(-pi * tn) – even function"""
    for tn in [0.1, 0.5, 1.0, 2.3]:
        assert abs(math.cos(math.pi * tn) - math.cos(-math.pi * tn)) < 1e-15
def test_Um_negative_tn_growth():
    """For tn < 0 with cos < 0, Um exponent becomes positive (growing)"""
    gamma = 0.0001; t = 100.0; tn = -1.0
    cos_val = math.cos(math.pi * tn) # = -1 for tn=-1
    Um_factor = 1.0 - math.exp(-gamma * t * cos_val)
    # -gamma * t * (-1) = +gamma * t → exp grows > 1 → factor becomes negative (growing regime)
    assert math.exp(-gamma * t * cos_val) > 1.0, "Negative tn should yield exp > 1"
def test_Ug1_zero_at_halfcycle():
    """Ug1 = 0 at tn = 0.5 (pi/2 null crossing)"""
    tn = 0.5
    cos_val = math.cos(math.pi * tn)
    assert abs(cos_val) < 1e-15, f"cos(pi/2) must be 0, got {cos_val}"
```

